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$$f(x, y) = x^2 + 2y^2 - 4x - 8y - 5$$

$$\frac{\partial f}{\partial x} = 2x - 4 \Leftrightarrow x = 2$$

$$\frac{\partial f}{\partial y} = 4y - 8 \Leftrightarrow y = 2$$

$$H(x, y) = \begin{vmatrix} 2 & 0 \\ 0 & 4 \end{vmatrix} = 8$$

$$\Rightarrow H(2, 2) = 8 > 0$$

$$\Rightarrow \frac{\partial^2 f}{\partial x^2}, \frac{\partial^2 f}{\partial y^2} > 0 \Rightarrow (2, 2) \text{ is een minimum}$$

References